Towards a Taxonomy of Sustainability Requirements for Software Design

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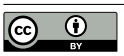
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Abstract

Contemporary software development is driven by two major imperatives: (i) the adoption of rapidly advancing technologies such as artificial intelligence, the Internet of Things (IoT), and cloud computing, and (ii) the adherence to global sustainability goals that encompass environmental, economic, and social dimensions. The first imperative both enables and challenges the achievement of sustainability goals. Current state-of-the-art research indicates that understanding sustainability concepts and their application in software development remains in an early stage. Sustainable software development methodologies (such as lean) and government legal mandates are effective in producing less carbon footprint and being a more socially inclusive team during the development process. However, a critical question remains: are the resulting software product features sustainable? A major limitation lies in the absence of well-defined artifacts that capture granular sustainability requirements for software systems. This research work addressed this limitation by conducting a systematic literature review and developing a comprehensive taxonomy of sustainability requirements for software development. It identified the categories or issues that must be addressed with different (environmental, economic, social, technical) dimensions of sustainability. Furthermore, the study contributes by identifying the correlations both positive and negative among these sustainability categories.

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1 Introduction

Software inherently contributes to global sustainability concerns [20], [12]. Its impact spans the entire life cycle, beginning with requirements engineering and extending through deployment and usage. The sustainability footprint of a software system is largely dependent on its features and design choices, making sustainable software design essential to minimize negative effects. There are two main directions for achieving this [20]. The first emphasizes the adherence to benchmark practices and guidelines, such as the Karlskrona Manifesto for green software engineering [5]. The second involves a deeper analysis of the specific software system under development to identify and address sustainability concerns. These two approaches can complement each other. The focus of this research lies on the second direction.

The existing literature provides substantial evidence [4, 7–9] that sustainability concerns need to be addressed from the requirements engineering phase. Although organizations realize this need, they lack concrete support and knowledge. The question that arises, *What are the sustainability requirements (SRs) for a system to be built?* Many research contributions [8, 25, 30] have already been made, formulating SRs that are necessary. These contributions are often specific to a certain sustainability dimension [1, 14, 31] or to application domains [2, 11]. This existing state-of-the-art can form a large database of SRs across various domains. However, there is a lack of systematization of these requirements that could serve as a reference for the software engineering community. This research work extracts and organizes these data into a comprehensive taxonomy. The taxonomy is designed to assist both software developers and researchers in formulating sustainable software development approaches. The designed taxonomy primarily contains SRs that may be checked against a software product in development. The main contributions of this work are as follows.

(1) A systematic literature review is conducted to create a taxonomy of sustainability requirements.

(2) A comprehensive set of categories for the four dimensions of sustainability (environmental, technical, social, and economic) is identified. These categories are defined in conjunction with their metrics and measures.
(3) A derivation of correlations among different sustainability categories.

The structure of the rest of the paper is as follows. Section 2 elaborates on the existing state-of-the-art. Section 3 illustrates the methodology adopted for selecting artifacts from the state-of-the-art. Section 4 describes the taxonomy construction process and categories identified. Section 5 describes how the correlation among sustainability categories is determined. Section 6 highlights some validity concerns and Section 7 concludes.

2 State of the Art

This section illustrates recent research proposals made to integrate sustainability requirements within the software development process. ShapeRE [28] is a theoretical framework designed for iterative identification and representation of sustainability requirements. Researchers [27] have assessed how Linguistic-Relativity Theory impacts identification of SRs. The authors have experimented with 101 participants, and findings show that the associating dimensions with requirements guides them in correctly identifying SRs. Researchers [29] have proposed a goal-based scenario and a feature modeling approach to decompose sustainability requirements. This approach enables the assessment of sustainability risk analysis. An empirical study [3] investigates sociocultural sustainability factors for an e-learning system. These results provide gender based social sustainability concerns recorded from 177 participants. A study conducted by the authors in [22] has identified key sustainability dimensions that are being considered within the software engineering process. Their findings show that the characteristics perceived to be the most significant were security, usability, reliability, maintainability, extensibility, and portability. But concerns like computer ethics (e.g. privacy and trust), functionality, efficiency, and re-usability have little attention. SuSAF [21] is a question-based requirements engineering framework that helps stakeholders systematically explore a software system's potential sustainability impacts across social, individual, environmental, economic, and technical dimensions. Similarly the Sustainability Assessment Framework (SAF) Toolkit [10], a set of modeling and assessment instruments, including the Decision Map and Sustainability-Quality (SQ) model, that help software architects systematically reason about sustainability concerns, trade-offs, and long-term impacts across multiple dimensions. A survey conducted in [13] have provided a comprehensive list of non-functional requirements (NFRs) and their association with four sustainability dimensions (environmental, economic, social, and technical). Researchers in [26] have listed sustainability requirements for healthcare applications in smart domains. It covers five dimensions of SRs-social, economic, individual, environmental, and technical.

The current state-of-the-art methodologies aim to integrate sustainability concerns within the development process. However, SRs identified are often only specific to certain domains or are framed in higher abstraction.

3 Methodology

This section presents the methodology used to construct the sustainability requirements taxonomy. Figure 1 shows the step-by-step process followed to identify and select relevant research works through a systematic literature review (SLR). The following sections provide a detailed explanation of each step.

3.1 Establish Research Questions

The formulation of a set of well-defined research questions forms the basis of a focused and effective SLR. These research questions are refined iteratively based on preliminary readings and understanding of the existing state of the art.

RQ-1 How are sustainability requirements defined and characterized across different domains (e.g., stand-alone software, cloud computing, smart systems, healthcare)?
RQ-2 What are the most frequently identified categories of sustainability requirements for different dimensions in the existing state of the art (e.g., inclusivity for social dimension, energy consumption for environmental dimension)?
RQ-3 What commonalities and divergences of sustainability requirements across different domains can be identified and recorded within a generic comprehensive taxonomy?

RQ-1 is intended to understand how sustainability requirements are defined for software systems in general and also to consider different possible application domains. *RQ-2* is intended to identify how much the different dimensions of SRs are explored to design software systems and what categories within each dimension are considered. *RQ-3* requires comparing SRs defined for different domains, understanding the overlaps among them, and also if they are too specific for a particular domain. This step ensures that the taxonomy does not contain redundancies in the captured list of SRs. The search strategy used to answer these research questions directly influences the granularity and comprehensiveness of the taxonomy.

3.2 Review Process

The details of the review process are as follows.

Selection of Databases

The authors examined the main digital libraries that contain peer-reviewed research in engineering, information systems, and software applications. The study focused exclusively on qualitative data sources from the literature. Qualitative data were collected from peer-reviewed journals, conference proceedings, and book chapters to identify SRs. The specific databases searched include-Scopus, ACM Digital Library, IEEE Xplore and wiley Online Library.

Search Strings and Database Search

This is the third and fourth step, where the search strings are defined and the search is conducted at different levels. The SLR search strategy was mainly adopted from the report of Kitchenham and Charters [19]. The main concern of this process is the creation of a search string. In the SLR process, three different types of search strings were created and the search was performed at three different levels-

(1) The first level involved a broad and generic search using the keywords "sustainability" and "requirements" in the selected

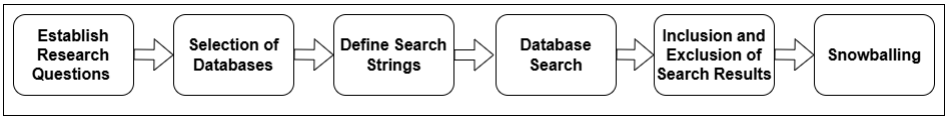


Figure 1: Steps of Systematic Literature Review

databases. The Boolean operators "AND" and "OR" were applied to include synonyms, spelling variations, and related terms. This initial search aimed to identify sustainability requirements (SR) associated with software or information systems. However, most of the studies retrieved primarily emphasized the environmental dimension of sustainability.

(2) The second level of search included more specific dimensions of sustainability. The search was extended to the next level by including terms related to different dimensions of sustainability, such as "Social", "Technical", and "Economic". As a result, we obtained research articles that focused primarily on specific dimensions of sustainability for software and information systems. The environmental dimension was excluded at this level to avoid redundancy with the results of the first search phase.

(3) The third level of search focused on specific application domains such as healthcare, Smart Home, E-learning, cloud and IOT systems, robotic Systems. This step was crucial since many application-oriented studies on sustainability requirements are indexed under domain-specific categories and may not appear in generic searches. However, this search was restricted to selected domains due to the time-bound nature of the systematic review.

Table 1 documents the search strings used and the corresponding number of retrieved records. The primary search employed the string in row 1, followed by secondary searches using the strings in rows 2–4. In case of domain-specific studies, these strings were concatenated with domain names, with examples of third-level search strings shown in rows 5–8. This keyword-based strategy (rows 1–8) supported RQ-1 and RQ-3, and enabled identification of sustainability categories (RQ-2) and associated metrics. As not all studies reported suitable metrics, an additional search using the final row of Table 1 was conducted in digital databases to identify metrics and measures for sustainability concerns across dimensions.

Inclusion and Exclusion of Search Results

In this step, specific criteria were defined to select relevant studies from the search results. Inclusion and exclusion criteria were manually applied to ensure that only works that directly address the research questions were considered. The first inclusion criteria was to select the first 100 records found for each search query. These records are then refined as follows-

(1) Peer-reviewed research works published in journals, conferences, and workshops.
(2) Research works that are written in English.
(3) Research works in the domain of computer science or software engineering.
(4) Studies in which SRs for any software system are explicitly mentioned in the title and abstract.
(5) Studies that explicitly mention sustainability frameworks, principles, or catalogs. These studies help, in particular, to identify categories and metrics.

The exclusion criteria are applied as follows.

(1) Studies published before 2010. This is because we find that in last 15 years there are a significant amount of literature that can be considered for this work.
(2) Studies that discussed sustainability in general but did not explicitly address requirements, or detailed definitions.
(3) Non-peer-reviewed studies like tutorials, posters, and other such articles.

Studies meeting the inclusion and exclusion criteria were selected from the database search. Table 1 reports the number of records retrieved per database and search string, with the top 100 records initially selected (columns 6-9). The search covered five domains—healthcare, smart home systems, cloud and IoT systems, and robotic systems to support domain-specific analysis; the total records across these domains are summarized in Row 10. Overall, 3,818 articles were retrieved and screened, of which 105 relevant studies were retained to construct the taxonomy. Duplicate records across databases were consolidated so that each study was included only once in the final dataset.

Snowballing

In the last step, snowballing is applied to consider the reference list in the selected research works. Citations made to selected research works are also considered. Inclusion and exclusion criteria are also applied to these referenced research works. The snowballing is performed only once, based on the works selected in the previous step.

4 The Taxonomy

The taxonomy was constructed after methodically analyzing each of the selected research works. The steps applied for taxonomy construction are as follows.

(1) The authors have divided the total number of selected research papers equally among themselves.
(2) The authors have read each paper to identify the objectives and main contributions -(in terms of SRs, identified categories and metrics).
(3) After initial analysis of each paper, the author recorded their findings and discussed them together.
(4) The initial findings have shown that some of the works have directly listed out SRs or goals. However, for other works, they have informally discussed sustainability requirements, but these are not definitive.
(5) We filtered out the studies that discussed SRs only in an informal or unstructured manner. These included 27% of the total 105 works selected. These selected works were then analyzed using AI tools with a structured prompt[1] designed

[1]https://docs.claude.com/en/docs/build-with-claude/prompt-engineering/overview

Table 1: Database Search Strings and Record Selection

Search String	Number of Records in each Database Found				Number of Records Selected				Total Records	After applying inclusion and exclusion criteria
	Scopus	IEEE Xplore Digital Library	ACM Digital Library	Wiley Online Library	Scopus	IEEE Xplore Digital Library	ACM Digital Library	Wiley Online Library		
"sustainability requirement" OR "sustainable requirement" OR "sustainability principle" OR "sustainable principle"	2863	18	15822	110	100	18	100	100	318	37
"social sustainability requirement" OR "social sustainable requirement" OR"social sustainability principle" OR "social sustainable principle"	886	1850	22,285	4735	100	100	100	100	400	15
"technical sustainability requirement" OR "technical sustainable requirement" OR "technical sustainability principle" OR "technical sustainable principle"	170	2241	33,088	4735	100	100	100	100	400	10
"economic sustainability requirement" OR "economic sustainable requirement" OR "economic sustainability principle" OR "economic sustainable principle"	6321	2897	14807	0	100	100	100	0	300	11
Search string of first 4 rows combined with Domains- Healthcare, Smart Home, E-learning, Cloud and IOT systems, robotic Systems	8059	368	7071	2593	500	500	500	500	2000	25
"sustainability metrics" OR "sustainability measure" OR "sustainability measurement"	8059	368	7071	2593	100	100	100	100	400	7
Total									3818	105

to extract potential SRs documented in the manuscript. The structured prompt used in this process is provided in the gitHub repository. The prompt was executed for both the AI tools (ChatGPT [12] and Gemini [15]) in case of each filtered study.

(6) The results obtained from the two AI tools are manually assessed and validated by the four co-authors (responsible for the particular manuscript) sharingly to avoid inclusion of hallucinated results.

(7) Finally, all the listed SRs are included in the taxonomy. Each SR is supported by its dimension and category.

Table 2 shows an abridged version of the SR taxonomy. The taxonomy is exhaustive and non-repetitive and it also contains requirements that may pertain to specific categories of systems (derived from research works of specific domains). The taxonomy [2] is open-source and the list of SR may be updated periodically.

4.1 Sustainability Categories and Metrics

In this research work, we have identified categories for four different sustainability dimensions-environmental, social, economic, and technical. The identified categories are mainly relevant to software and information systems. Some of the categories were identified from the research works selected to build the taxonomy. Other results were obtained from Table 1. These results were again subjected to inclusion and exclusion steps. Similar steps are applied here (as in taxonomy construction) to identify categories and metrics. Some of the metrics can be evaluated during the software development phase, while others can only be assessed after deployment. For example, within the social dimension, the social value of the category customer can be measured using the metric amount of positive feedback, which becomes available only once the system is in use.

We found that some categories are associated with multiple metrics, such as energy consumption, which includes indicators like runtime efficiency [18], memory usage [18], and algorithm efficiency [16]. Some metrics have well-defined formulas (e.g., software carbon intensity [23], development cost [17], cohesion and coupling [24]), whereas others rely on expert judgment due to the lack of precise quantification methods. The full list of referenced studies, categories, metrics, and measures is available in the GitHub repository[2]. Overall, 87 papers addressed RQ-1, 41 addressed RQ-2, and 31 addressed RQ-3.

[2]https://github.com/MCompRETools/SustainabilityTaxonomy

5 Correlation among Sustainability categories

Each sustainability dimension comprises multiple categories that influence one another [6]. While prior work [21] maps run-time and architectural NFRs to sustainability dimensions and defines some interrelations, sustainability extends beyond traditional NFR categories, and their correlations remain largely implicit. In this work, we employ a large language model (LLM)–based approach to derive a catalog of sustainability category correlations. A dataset comprising sustainability categories under each dimension, along with their definitions from the state-of-the-art, was first constructed. This dataset, together with a structured prompt (refer to github Repository[2]), was then provided as input to a LLM to generate a correlation matrix capturing potential interrelations among categories. The LLM-generated correlations were independently reviewed by each author, who recorded their agreement or disagreement. All feedback was subsequently consolidated into a single worksheet and discussed collectively by the four authors. A consensus-based voting mechanism is used and the most appropriate correlations were finalized and recorded. In result generation, LLMs GPT4 [12] and Gemini 2.5 Pro [15] have been used. We found that both LLMs have generated similar results and reasoning. The detailed correlations can be found in the github repository [2].

6 Threats to Validity

The first threat to validity concerns the completeness of the taxonomy across diverse software domains. While we considered generic literature, domain-specific SRs were explored for only a limited set of domains. However, the taxonomy is open-sourced and designed for iterative updates through versioning, and its application is inherently dependent on the specific software system being developed. A second concern relates to the selection of research works published from 2010 onwards. Our survey indicates that these studies build upon and synthesize earlier research while introducing new insights. Therefore, the selected state-of-the-art literature sufficiently captures the necessary foundational and contemporary knowledge. The third threat involves the use of LLMs for result generation. However, authors have manually checked the LLM generated responses to prevent documenting hallucinated results.

7 Conclusion

This research work systematically reviews the state of the art on SRs for software systems, resulting in (i) an open-source taxonomy of SRs and (ii) a set of categories and metrics across sustainability

Table 2: Sustainability Requirements Taxonomy (abriged)

Sustainability Requirement	Dimension	Category
Assessing software over a long term and consider decisions with this in mind.	Technical	Longevitivity
An appropriate low-energy color scheme should be chosen based on the monitor technology	Environmental	Energy Consumption
Appropriate tones should be chosen in a color gradient that depends on the particular technology of the monitor	Social	Social Engineering
Menus should present as few choices as possible to reduce the human cognitive process of decision-making, in line with Hick-Hyman Law	Technical	Feature Bloat
Ensuring software systems do not perpetuate or exacerbate social inequalities.	Social	Equitable Access
Ensuring systems promote fair market practices.	Economic	Affordability

dimensions. The taxonomy is designed for iterative extension and can support developers in identifying domain-specific sustainability concerns, while the identified categories and metrics enable quantitative assessment of sustainability goals. The fifth dimension of sustainability, which is individual, is not kept within the scope of this work. A further contribution is the identification of correlations among sustainability categories, highlighting positive and negative interdependencies across dimensions. These correlations can aid developers in managing trade-offs when designing sustainable software.

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